SUBHERO — ENTERPRISE INFRASTRUCTURE & BUSINESS ARCHITECTURE

SubHero — Technical Blueprint

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What SubHero Is

SubHero is a social media growth and digital services platform built for the Nigerian market. At its core, it gives individuals, creators, and small businesses affordable access to social media marketing tools — followers, views, likes, comments, and engagement across every major platform — alongside daily essentials like mobile data, airtime, and bill payments, all powered by a single wallet.

The platform is live at `subhero.ng`.

The Market Opportunity

Nigeria's creator economy is exploding. There are an estimated 20M+ active social media users building audiences on Instagram, TikTok, YouTube, and Twitter. Small businesses, musicians, skit makers, influencers, and even politicians need social proof to grow — but the existing tools are either overpriced, foreign (priced in USD), or outright scams.

On the other side, 150M+ Nigerians buy data and airtime every week, multiple times a week. That's recurring, habitual spending that happens regardless of economic conditions.

SubHero sits at the intersection of these two massive markets:

- **Social media growth** (high margin, fast-growing) — the headline product
- **Data, airtime, and bill payments** (high frequency, daily retention) — the engagement engine
- **Viral giveaways** (zero-cost acquisition) — the growth flywheel

No existing platform in Nigeria combines all three.

How It Works

Social Media Marketing (SMM) — The Core Product

SubHero offers **3,667 social media marketing services** across six platforms:

Platform	Services Available	Examples
Instagram	1,541	Followers, likes, views, comments, story views, reel plays, auto-engagement
TikTok	652	Followers, views, likes, shares, saves, live stream viewers
YouTube	654	Subscribers, watch hours, views, likes, comments, shorts views
Spotify	299	Plays, followers, monthly listeners, playlist adds, saves
Twitter/X	266	Followers, retweets, likes, impressions, poll votes, spaces listeners
Telegram	255	Channel members, group members, post views, reactions

How a typical SMM order works:

1. User selects a platform (e.g., Instagram), a service (e.g., "1,000 Real Followers"), and pastes their profile URL
2. SubHero validates the URL against an allowlist of legitimate social media domains (SSRF protection)
3. User confirms the Naira price and enters their 4-digit transaction PIN
4. Wallet is debited atomically — the order is dispatched to SubHero's fulfillment network via background task
5. Delivery begins within minutes. Progress is tracked and polled automatically every 30 minutes
6. User receives a branded receipt once the order completes

Why this is the money maker:

SMM services carry **40-60% gross margins**. SubHero sources at wholesale USD rates (fractions of a cent per 1,000 units), converts to Naira at markup, and sells in denominations that feel cheap to the end user. A Nigerian skit maker paying ₦1,200 for 1,000 Instagram followers doesn't think twice — it's less than two meals. SubHero's cost for that order is roughly ₦500-₦700.

Data, Airtime & Bills — The Retention Layer

These aren't side features — they're what keeps users opening SubHero every single day. Nigerians buy data 2-4 times per week. If they can do that in the same app where they boost their Instagram, they don't need to go anywhere else.

SubHero handles:

- **Mobile Data** — MTN, Airtel, Glo, 9mobile (SME, corporate, gifting plans)
- **Airtime Top-Up** — all networks, instant delivery
- **Cable TV** — DStv, GOtv, StarTimes subscriptions
- **Electricity** — prepaid meter tokens (all discos)
- **Exam PINs** — WAEC, NECO, NABTEB scratch cards

Every service runs through the same architecture: wallet debit → upstream API call → success or automatic refund.

Giveaway System — The Growth Engine

This is SubHero's built-in user acquisition machine, and it costs the company nothing to run.

The concept: Any SubHero user can host a giveaway — distributing free data or airtime to strangers — and SubHero handles everything from escrow to fulfillment. The host gets social clout. SubHero gets new users and earns margin on every single slot claimed.

How it works step-by-step:

1. A registered user creates a giveaway — picks data or airtime, selects a network, sets the number of slots (e.g., "20 people get 1GB MTN data each")
2. SubHero calculates the total cost and **debits it upfront into a locked escrow** — the funds are ring-fenced, not spendable
3. A unique short link (`subhero.ng/g/XXXX`) and a downloadable QR code are generated
4. The host shares the link on Twitter, Instagram stories, WhatsApp groups, university forums, or anywhere they want attention
5. **Anyone who clicks the link can claim a slot** — they don't need a SubHero account, they just enter their phone number
6. SubHero fulfills the data or airtime to the claimant's phone number in real time, drawing from the escrowed funds
7. Once all slots are claimed (or the giveaway expires), any unclaimed funds are automatically refunded to the host's wallet

   Why this is a strategic weapon:

   - **Zero customer acquisition cost.** SubHero doesn't pay for the data being given away — the host does. But SubHero earns the margin on every unit fulfilled. It's user-funded marketing where SubHero profits on every impression.
   - **Viral by design.** A single giveaway with 50 slots means 50 people interact with SubHero's brand, receive value from the platform, and see a branded claim success page that reads: *"Host your own giveaway at subhero.ng."* A downloadable receipt card with SubHero branding is generated for every giveaway — hosts share these as social proof.
   - **Conversion funnel built in.** The claim success page is a full conversion card targeting unregistered claimants — it shows SubHero's pricing, a testimonial, and a "Create Free Account" CTA. Every giveaway is a landing page that SubHero didn't have to build or pay to promote.
   - **Network effects.** As more users host giveaways, more non-users encounter SubHero. Some of those convert and eventually host their own giveaways. The loop compounds.
   - **Revenue on both sides.** SubHero earns margin when the host funds the giveaway (wallet top-up), earns margin on every data/airtime unit fulfilled during claims, and earns again when converted claimants start buying services themselves.

Growth math: If 100 users each host one giveaway per month with 20 slots, that's 2,000 brand impressions per month — from people who actually received something valuable. Even a 10% conversion rate means 200 new registered users per month from giveaways alone, at zero acquisition cost.

Wallet & Payments

Users fund their SubHero wallet through two channels:

- **Card/Transfer/USSD:** Squad (GTBank) processes the payment → real-time webhook → wallet credited in under 2 seconds
- **Dedicated Virtual Account:** One-time KYC with BVN → permanent bank account number provisioned → user transfers anytime from any banking app → auto-credited via webhook

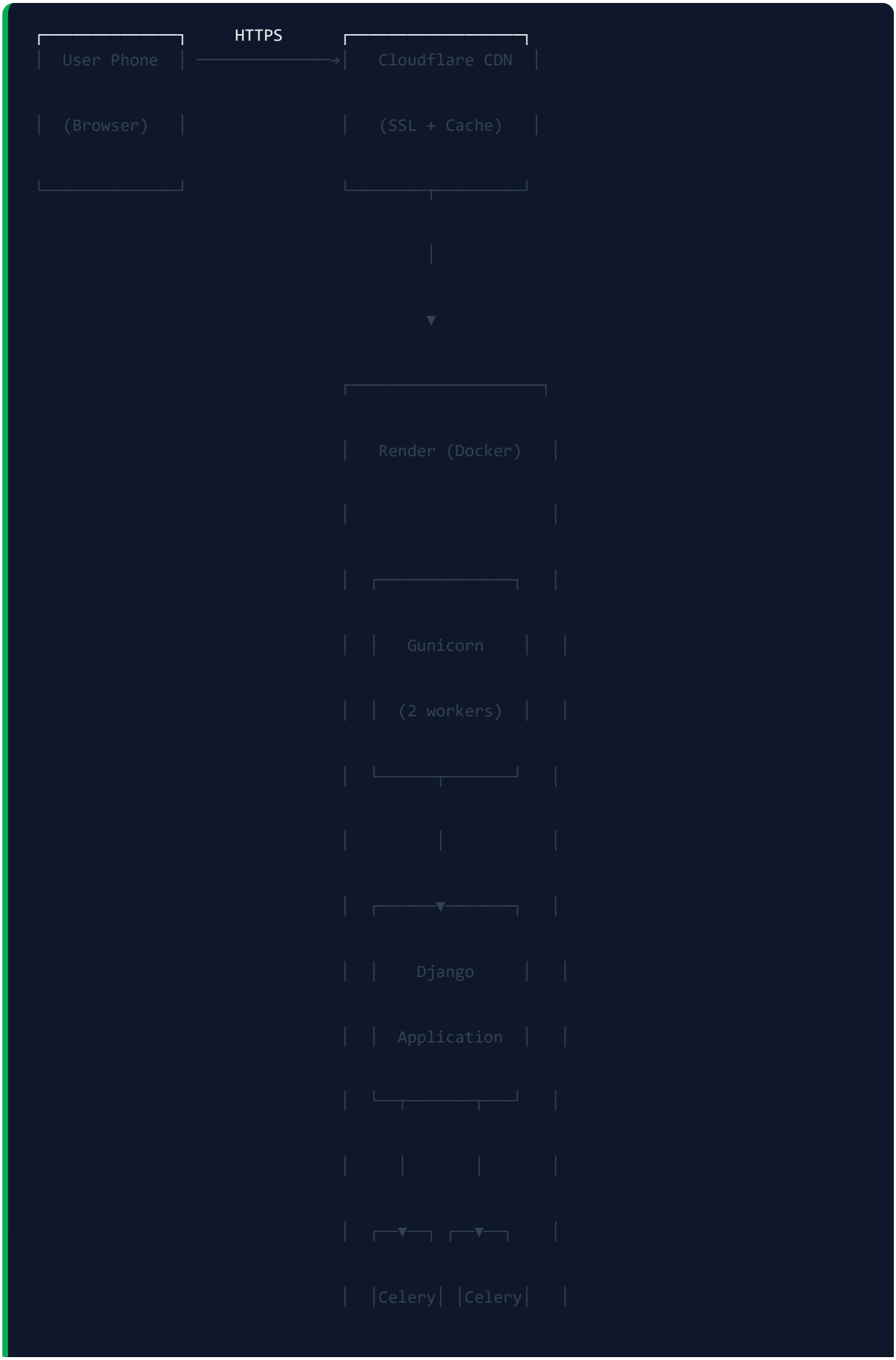
The wallet creates switching costs. Once a user has balance, they transact on SubHero instead of going to competitors who require card details on every purchase.

Architecture

Stack

Layer	Technology
Backend	Django 5.x (Python 3.12)
Database	PostgreSQL 16 (managed)
Cache / Queue Broker	Redis (managed, TLS)
Task Queue	Celery (worker + beat scheduler)
Frontend	Server-rendered templates + TailwindCSS + HTMX
Payments Gateway	Squad (GTBank subsidiary)
SMM Fulfillment	Proprietary wholesale fulfillment network
VTU Providers	Sabuss, CheapDataHub
SMS / OTP	Termii
Hosting	Render (Docker containers)
DNS / CDN	Cloudflare

System Diagram




|Worker| | Beat |
PostgreSQL
(Managed)
• Users
• Wallets
• SMM Orders
• SMM Catalogue
• VTU Orders
• Giveaways
• Webhook logs
Redis
(Managed TLS)
• Sessions
• Rate limits
• Celery broker
• Idempotency
keys
External APIs (outbound):
• SMM Network → Wholesale fulfillment (3,667 services, 6 platforms)
• Squad (GTBank) → Payment initiation, webhooks, virtual accounts

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• VTU Providers  → Data, airtime, cable TV, electricity, exam PINs

• Termii         → SMS OTP delivery
```

SMM Order Lifecycle

```
User selects service + pastes URL + enters PIN
    │
    ▼
Django view: validates URL domain against allowlist
    │
    ▼
PostgreSQL: BEGIN TRANSACTION
  → SELECT wallet FOR UPDATE (row-level lock)
  → Check balance ≥ price
  → Debit wallet
  → Create WalletTransaction
  → Create SMMOrder (status=PENDING)
COMMIT
    │
    ▼
Celery task dispatched: process_smm_order
    │
    ▼
Call SMM Fulfillment API → dispatch order (service, target URL, quantity)
    │
    ├── Success → Store provider_order_id, status → PROCESSING
```

```
│             → Celery beat polls status every 30 min
│             → Provider confirms completion → status → SUCCESS
│
└── Failure → Celery task: refund_smm_order
            → Credit wallet back, status → FAILED
```

Security

- **Argon2** password and PIN hashing
- **HSTS** with 1-year max-age + preload
- **UUID primary keys** on all models (prevents ID enumeration)
- **Row-level locking** (SELECT FOR UPDATE) on every wallet mutation
- **SSRF protection** — SMM target URLs validated against a strict allowlist of social media domains
- **Webhook signature verification** (HMAC-SHA512) on all payment notifications
- **Rate limiting** on login, OTP, and payment endpoints
- **PIN lockout** after 3 failed attempts, **Login lockout** after 5 failed attempts
- **IP logging** on every authentication event and transaction

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Revenue Model

SubHero earns on the spread between wholesale cost and retail price on every transaction.

SMM Revenue (Primary)

Service	Wholesale Cost	SubHero Price	Margin
1K Instagram Followers	~₦500	₦1,200	₦700 (58%)
10K TikTok Views	~₦150	₦400	₦250 (62%)
1K YouTube Views	~₦600	₦1,500	₦900 (60%)
1K Twitter Followers	~₦800	₦1,800	₦1,000 (55%)
1K Spotify Plays	~₦300	₦700	₦400 (57%)
1K Telegram Members	~₦700	₦1,500	₦800 (53%)

Average SMM gross margin: **50-60%**

VTU & Bills Revenue (Secondary)

Service	Wholesale Cost	SubHero Price	Margin
MTN 1GB Data	₦290	₦350	₦60 (20%)
Airtel 2GB Data	₦530	₦650	₦120 (22%)
₦500 Airtime	₦485	₦500	₦15 (3%)
DStv Compact	₦12,400	₦12,500	₦100 (0.8%)

Revenue Scaling Projections

Metric	Month 3	Month 6	Month 12
Registered Users	2,000	15,000	80,000
Monthly Active Users	800	6,000	35,000
SMM Orders/Month	1,200	12,000	75,000
Avg. SMM Margin/Order	₦600	₦650	₦700
SMM Gross Margin	**₦720K**	**₦7.8M**	**₦52.5M**
VTU/Bills Orders/Month	5,000	40,000	300,000
Avg. VTU Margin/Order	₦55	₦60	₦65
VTU Gross Margin	**₦275K**	**₦2.4M**	**₦19.5M**
Total Monthly Gross	**₦995K**	**₦10.2M**	**₦72M**

SMM contributes **~72% of total gross margin** despite being lower in transaction volume than VTU. The combination of high margins and growing Nigerian creator demand makes it the primary revenue engine.

How It Scales

Current Architecture (up to ~50K MAU)

- Single Docker container: Gunicorn (2 workers) + Celery worker + Celery beat
- Managed PostgreSQL with connection pooling
- Managed Redis for sessions, cache, and task queue
- Automated catalogue sync — Celery beat refreshes SMM service catalogue daily at 3 AM, VTU plans every 6 hours

Phase 2 (~50K-500K MAU)

- Dedicated Celery worker containers for SMM order processing (horizontally scaled)
- PgBouncer connection pooling in front of PostgreSQL
- Read replicas for analytics and admin dashboards
- Add 2-3 backup SMM providers for failover and geographic reach
- CDN tuning on Cloudflare

Phase 3 (500K+ MAU)

- Kubernetes deployment with auto-scaling pod groups
- Separate read/write database routing
- Event-driven order processing pipeline
- Dedicated fraud scoring system
- Multi-region deployment

What Doesn't Need to Change

The database schema, business logic, and provider abstraction layers are built to work at any scale. PostgreSQL handles millions of rows. Redis-backed Celery scales by adding workers. The codebase already uses:

- Row-level database locks (no race conditions)
- Idempotency keys on every financial operation
- Atomic transactions wrapping every debit-order pair
- Background task architecture for all SMM fulfillment

Competitive Advantages

1. **Largest SMM Catalogue in Naira.** 3,667 services across 6 platforms, all priced in Naira. Most competitors either price in USD (alienating the mass market) or offer a handful of services. SubHero's breadth means a user can grow on Instagram, TikTok, YouTube, Twitter, Spotify, and Telegram from one dashboard.
    1. **The Giveaway Viral Loop.** No competitor has this. Every giveaway is user-funded marketing that SubHero earns margin on. The host pays, SubHero profits, and 10-50 new people discover the platform per event — zero customer acquisition cost.
        1. **Daily Retention Through Utility.** Data and airtime purchases happen 2-4× per week. By combining SMM with everyday telecom services, SubHero is an app users open daily — not just when they need followers. That daily engagement drives organic discovery of SMM services.
            1. **Full Wallet Model.** Wallet balance creates switching costs. Users fund once, spend across services. This drives higher lifetime value than one-off payment flows.
                1. **Mobile-First UX.** Built for Nigerian phone screens. Dark mode, smooth animations, HTMX-driven partial page updates, branded receipts. Noticeably better than the clunky interfaces dominating the current market.

Current Status

- **Live in production** at `subhero.ng`
- Processing real transactions with real money through Squad (GTBank)
- Connected to live SMM fulfillment network
- Connected to live VTU providers for all major Nigerian networks
- Full admin dashboard — operations, user management, financial monitoring, system logs
- Automated service catalogue syncing across all providers

What Investment Enables

1. **Creator Marketing & Growth** — Influencer partnerships, campus ambassador programs, skit-maker sponsorships, and social media campaigns targeting SMM buyers directly
2. **Native Android App** — Push notifications for order delivery updates, offline-first UX, better retention (80%+ of Nigerian smartphones are Android)
3. **SMM Provider Expansion** — Integrate 2-3 additional SMM fulfillment networks for faster delivery, broader service coverage, and automatic failover
4. **Compliance Infrastructure** — CBN fintech licensing, NIBSS integration, formal KYC/AML
5. **Team** — Customer support, a mobile engineer, and a dedicated growth hire

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